Sepehr Khodadadi Hosseinabadi

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Research Interests

Trustworthy machine learning, machine unlearning, data privacy, model evaluation and auditing, and unlearning in federated settings; privacy-critical AI for medical imaging.

Education

2024 – Present **M.Sc. in Data Science and Engineering (AI track)**, University of Genoa, Italy
GPA: 30.2/30 (lode counted as 31)
Thesis: *Machine Unlearning for Model Debiasing* (in progress; advisors: Prof. V. P. Pastore (MaLGa), Prof. E. Tartaglione (Télécom Paris))
Relevant Coursework: Trustworthy AI (30/30; adversarial robustness, evasion & poisoning attacks, LLM security, EU AI Act), Deep Learning (Transformers, Diffusion Models, representation learning), Machine Learning, Computational Vision (segmentation, pose estimation, 3D reconstruction), Natural Language Processing & Symbolic and Distributed Artificial Intelligence (multi-agent systems, Prolog), High-Performance Computing (CUDA), Digital Signal & Image Processing

2019 – 2024 **B.Sc. in Computer Engineering**, Isfahan University of Technology, Iran
GPA: 3.2/4.0; Last 60 credits GPA: 3.55/4.0
Thesis: *Exploring the Efficacy of Local Search Methods for Enhancing the Performance of the BRADO algorithm.* **Grade: 20/20**

Publications

- S. Khodadadi Hosseinabadi, E. Iakovleva, E. Tartaglione, and V. P. Pastore (2026). *TRAIL: Navigating the Jungle of Machine Unlearning Evaluation*. Accepted (oral presentation), ECCV 2026 Workshop and Challenge on Unlearning and Model Editing (U&ME).
- S. Khodadadi Hosseinabadi et al. (2026). *Forget-Conditioned Subnetworks for Machine Unlearning.* Manuscript in preparation (target: ICLR 2027).
- S. Ramezani Jolfaei and S. Khodadadi Hossein Abadi (2025). *Application of the Brain Drain Optimization Algorithm to the N-Queens Problem*. arXiv:2504.18953.

Research Experience

M.Sc. Thesis Research: Machine Unlearning Feb 2026 – Present
MaLGa (Machine Learning Genoa Center), in collaboration with Télécom Paris

- Developed *forget-conditioned subnetworks*: a structural approach to machine unlearning.
- Delivered the SOTA results and evaluated under membership-inference (MIA) and domain-specific metrics, e.g., relearning attacks.
- Devised the critical idea of a two-phase fine-tuning schedule and adding the critical ∼5 retain-accuracy points, validated across seeds.
- Ran the control suggested by the external review of my thesis, retraining only the classifier head: it matched a retrained model on every accuracy metric, yet ten forget images brought the class back.
- Reran the relearning comparisons in the competing methods' protocol, so that results are comparable with the reported ones.
- Replicated a published relearning attack and mapped the method's limits under it: forgetting survives 5% forget-data mixing (recovering only ∼11%, near random for ten classes) but collapses at 10%.

TRAIL: Reproducible Evaluation of Machine Unlearning May 2026 – Jul 2026

First-author paper, accepted as an oral presentation at the ECCV 2026 Workshop and Challenge on Unlearning and Model Editing (U&ME)

- Built **TRAIL**, a protocol-only, multi-metric evaluation library that runs on user-supplied checkpoints and data loaders and produces **provenance-stamped reports** with unified metric definitions.
- Built the per-example membership-inference tier of the library (**U-LiRA**), with non-members drawn from held-out examples of the forgotten class.
- Built the **metric-correlation analysis**: a Spearman map over four unlearning methods and three seeds on CIFAR-10/100 (public and lab-trained checkpoints) of a registry of 45 metrics across three axes, 41 of them measured, finding that the axes measure largely distinct things.

Teaching Experience

University Teaching Assistant, Database Laboratory Sep 2022 – Sep 2024
Isfahan University of Technology, Isfahan, Iran

- Led weekly lab sessions on **SQL Server** for **50+** students across four semesters; co-developed the lab curriculum with faculty.

Awards & Honors

- **Reviewer**, ECCV 2026 FAILED Workshop (Model Debiasing).

Technical Skills

Trustworthy ML	Trustworthy machine learning, data privacy | machine unlearning, model evaluation and auditing, membership-inference evaluation, relearning-attack protocols, unlearning benchmarking | Iterative Magnitude Pruning, LiRA, U-LiRA
Machine Learning	Deep learning | experiment management, seeded deterministic pipelines, provenance-stamped reporting | PyTorch, TensorFlow/Keras, Weights & Biases, Hydra
Systems and HPC	High-performance computing | HPC clusters
Programming Languages	Python, C/C++, CUDA

Research Projects

Multi-Agent Clinical Decision Support for Lung-Nodule Assessment | **Python, Prolog (PySwip), scispaCy**

- Built a BDI multi-agent system fusing imaging agents (chest X-ray CNNs via TorchXRayVision, rule-based Lung-RADS) and report-text agents (dependency-anchored frame extraction, NegEx-style negation and uncertainty detection) under a Prolog weighted-voting consensus with conflict-resolution rules and explanation generation.
- Separated aleatory from epistemic uncertainty and computed per-case dynamic agent weights from information richness: graded, interpretable confidence instead of categorical labels.

Retinal Blood Vessel Segmentation with Attention U-Net | **TensorFlow/Keras, PyTorch**

- Built an Attention U-Net with a hybrid Dice+Focal loss, reaching **96.4% accuracy** and **77.9% F1** (all-pixel; original course project in TensorFlow/Keras).
- Cut each training image into overlapping 128×128 windows, turning 20 images into 63,800 patches that keep every vessel at its true size, and averaged the predictions of overlapping windows to segment full images.
- Audited the model's failure modes via ground-truth/prediction overlays, localizing systematic errors on thin, low-contrast vessels which led to the analysis that motivated the hybrid loss design.
- Re-implemented the pipeline in PyTorch: the version published in the linked repository.

Interpretable Vessel Segmentation with Anisotropic Directional LoG (ALoG) | **Python, OpenCV**

- Built a training-free, interpretable-by-design classical pipeline for DRIVE vessel segmentation (CLAHE preprocessing; multi-scale, multi-orientation anisotropic LoG filtering with grid-searched filter parameters), a data-efficient

counterpart to the Attention U-Net for studying the accuracy–interpretability trade-off.

3D Geological Modeling and Optimization | **Trimesh, Open3D**

- Designed a mesh slicing, voxelization, and optimization pipeline reaching **95% material utilization**, with real-time Jupyter/Plotly visualization and live efficiency metrics.

Languages

English (IELTS: 8)	Listening: C2	Reading: C2	Speaking: C1	Writing: C1
Italian	Listening: B1	Reading: B1	Speaking: A2	Writing: B1
German	Self-study in progress.			
Persian (Farsi)	Mother tongue.			

References

Prof. Vito Paolo Pastore, MaLGa / DIBRIS, University of Genoa vito.paolo.pastore@unige.it

Volunteering

Co-organized an IELTS-preparation class to strengthen English skills among peers at Isfahan University of Technology.